

so i have an SE project, ill just give it to you

https://github.com/SETAPESU26/53_arm_wrestle

go through this and lmk what else you need

Pulling up the repo now.

Read a page

Read a page

Got the README. Now the actual code.

Read 2 pages

Read 2 pages

Read all of it. It's small: `main.py` (loop), `game/game_engine.py` (everything), and the README.

What's actually in the lab

- Task 1: the sign bug. Both arrow keys do `arm_position += 4.2`, so they should be `-= 4.2`.
- Task 2: AI surge/exhaust cycle in `update()`.
- Task 3: exhaustion indicator.
- Task 4 is a copy-paste duplicate of Task 3 in the README, so there are really only two features despite "three optional features." Say so in your submission instead of silently skipping it.

Problems the README doesn't mention

- Exhaustion basically can't happen. Each press costs 2 stamina, but regen is 0.8 per frame, which is 48/sec at 60 FPS. You'd need over 24 presses/sec to drain it. Even if you hit it, regen pushes you above 10 in one frame, so the lockout lasts about 16ms. A Task 3 indicator would never show up in your "after" video unless we retune the drain/regen numbers.
- Balance is broken after the fix. Player does roughly 40 units/sec, the AI about 14/sec, so you win in about 3.5 seconds. The surge mechanic needs to be strong enough to matter.
- Threshold mismatch. README says "below 10", code uses `<= 10`, and the bar turns red at 25. Pick one and make the indicator consistent.
- `reset()` will miss the new AI state. Surge/exhaust timers need resetting there too, or rematches start mid-surge.

What I need from you

1. Do you want me to write the full patched `game_engine.py` for you to review and test, or go task by task? The rubric is the chat history, so going task by task with you poking at it probably looks better.
2. Which indicator for Task 3: flashing red bar, "EXHAUSTED!" label, arm tremble, or a combo?
3. Any rubric or marking criteria beyond the README? Same for any TA constraints on how much the LLM should do.
4. Is pygame set up locally? I can't record gameplay videos here, so you'll need to capture the before/after clips yourself.

Keep it all in this one chat. You need to submit the full history link.

we also got some extra info

Lab 4: VibeCoding Objective Use Vibe Coding tools to fix the broken code and add features to the same. Learn how to prompt to get the job done in under three to four attempts. Clone a repo assigned to you, make changes and push to your repo in your personal account. **THIS IS AN INDIVIDUAL ASSIGNMENT. WORK INDIVIDUALLY!!** Duration 60 minutes Software Requirements • Access to Vibe Coding tool. (ChatGPT, Gemini, Claude etc) Instructions: 1.

Go to the repo assigned to you 2. Go through the ReadMe file to understand the deliverables 3. Clone or Fork the repo 4. Run the Python based code (A game/ A Puzzle) 5. Record 10 seconds of video before making any changes. 6. Write prompts and fix the broken code 7. Write prompts to add all the features listed in the ReadMe file of the repo one by one. 8. Record another 10 seconds of video again after making all the changes. 9. Push the code to your repo. 10. Have Separate Commits for each task 11. Do not raise PR to the main repo (SETAPESU26) Deliverables: Upload the following to your Lab repo under Lab-4 folder, a. Videos (before and after) b. Updated Code c. Chat history exported as a doc/pdf this is the handout we got, i need you to carefully guide me through and every step

Good, the handout changes a few things. Here's the full flow, in the order you'll do it.

Rules that matter

- Individual work, fork into your account, never open a PR to SETAPESU26 (GitHub will nudge you to; ignore it).
- One commit per task, so commit after each task, not at the end.
- They grade prompt efficiency: each task should take 3-4 prompts max. Paste the exact README task text, say which file to edit, and ask for only the changed code. Vague prompts burn your attempts.
- Deliverables go in a Lab-4/ folder in your repo: before/after videos, updated code, chat history as doc/pdf.

Phase 0: Setup

1. Fork SETAPESU26/53_arm_wrestle to your account (Fork button, top right).
2. `git clone https://github.com/<your-username>/53_arm_wrestle.git`, then `cd 53_arm_wrestle`
3. `pip install pygame`, then `python main.py`
4. Confirm the window opens and the arrow keys respond.

Phase 1: "Before" video

5. Record 10 seconds of you mashing Left/Right alternately so the hands visibly drift toward the computer's side. That's the bug. Don't leave it idle, because the AI alone also pushes the arm that way and it won't show the bug.
 - Windows: Win+G (Game Bar). Mac: Cmd+Shift+5.
6. Save it as `before.mp4`. Keep it out of the repo for now.

Phase 2: Task 1 (bug fix)

It's two lines in `handle_event`. Both `self.arm_position += 4.2` lines become `-= 4.2`.

Still run it through me for the chat record: tell me the task and I'll confirm and patch.
Then:

- Run the game and check that mashing now moves the hands toward the player side and you can win.
- `git add game/game_engine.py`
- `git commit -m "Task 1: fix inverted arm push bug"`

Phase 3: Task 2 (AI surge)

Send me the task in your own words, and I'll write the surge/exhausted cycle. You test it, then commit as `Task 2: add AI surge/exhaustion mechanic`.

Phase 4: Task 3 (exhaustion indicator)

Same loop. This commit also needs the stamina rebalance I flagged, otherwise your video can't show the indicator. Tasks 3 and 4 are identical, so cover both in one commit: `Task 3/4: add exhaustion warning indicator (Task 4 duplicate of Task 3 in README)`.

Phase 5: "After" video

Record 10 seconds showing all three: pushing works, a surge visibly happens, and the exhaustion warning triggers. You'll have to mash hard to reach it. Save it as `after.mp4`.

Phase 6: Deliverables

1. Make a `Lab-4/` folder containing `before.mp4`, `after.mp4`, a copy of the updated `game_engine.py` and `main.py`, and the chat export.

2. Export this chat as PDF. Browser print-to-PDF on the chat page works. Do it last, so it contains the whole conversation.
3. `git add Lab-4`, then `git commit -m "Add Lab-4 deliverables"`
4. `git push`

Two questions before we start

1. Does your course expect a separate lab repo with a `Lab-4` folder, or is the fork fine? “Your Lab repo” in the handout is ambiguous, so check with your TA or a classmate. It changes where you push the final commit.
2. For the exhaustion indicator, which do you want: flashing red bar, “EXHAUSTED!” label, arm tremble, or a combo? A combo is more impressive in a 10-second video.

Do Phases 0 and 1 now and tell me when the before video’s done. Then we go straight into Task 1.

okay wait walk me through phase 0

Phase 0, one step at a time. Do each one and tell me if anything breaks.

Step 1: Check what you have installed

Open a terminal (Windows: search “PowerShell” or “cmd”. Mac: Terminal) and run:

```
git --version
```

```
python --version
```

- Git missing: install from git-scm.com, then reopen the terminal.
- Python needs to be 3.10 or higher. On Mac/Linux the command may be `python3 --version`. If so, use `python3` and `pip3` for everything below.

Step 2: Fork the repo

1. Log into GitHub with your own account (github.com/SuhanaAGhosh).
2. Go to github.com/SETAPESU26/53_arm_wrestle
3. Click Fork (top right), then Create fork on the next page. Leave the defaults.

4. You should land on `github.com/SuhanaAGhosh/53_arm_wrestle`. Check that the URL has your username, not SETAPESU26.

Step 3: Clone your fork

Pick a folder where you keep projects (Desktop is fine), then run:

```
cd Desktop
git clone https://github.com/SuhanaAGhosh/53_arm_wrestle.git

cd 53_arm_wrestle
```

If git asks you to sign in, use your GitHub login in the browser prompt. If it asks for a password in the terminal, that has to be a Personal Access Token, because GitHub no longer accepts account passwords there.

Run `dir` (Windows) or `ls` (Mac) and you should see `game`, `main.py`, and `README.md`.

Step 4: Install pygame

```
pip install pygame
```

If you get “pip not found”, try `python -m pip install pygame`.

Step 5: Run the game

```
python main.py
```

A dark window titled “Arm Wrestle Showdown” should open, with a brown table, two arms, and a stamina bar at the bottom. Click the window, then mash Left and Right alternately. You should see the hands slide toward the computer’s side (the bug), and eventually “COMPUTER WINS!” appears. Press R to rematch.

If you don’t see that, tell me what you do see.

Step 6: Set your git identity (if you’ve never used git on this machine)

```
git config --global user.name "Suhana"
```

```
git config --global user.email "your-github-email@example.com"
```

Use the email tied to your GitHub account, or your commits won't show up on your profile.

Tell me what OS you're on and whether the game window opened. Then Phase 1, the before video.

not getting `git --version`

That means git isn't installed, or the terminal can't find it. Quick fix:

Windows

1. Go to git-scm.com/download/win. The download should start automatically.
2. Run the installer and click Next through everything. The defaults are fine.
3. Close your terminal completely and open a new one. This is the step people skip. The old window won't see the new install.
4. Run `git --version` again. You should see something like `git version 2.x.x`.

Mac

Run `git --version`. If git isn't installed, macOS usually pops up a prompt to install the Command Line Tools. Click Install and wait a few minutes. Then try again.

What exactly did you see?

If it's still failing after the install, send me the exact error text:

- `'git' is not recognized...` means it isn't on PATH. Reopen the terminal, and if that doesn't work, reinstall and make sure the "Git from the command line" option is selected.
- Nothing at all, or a different message, means paste it here and I'll sort it out.

Also run `python --version` while you're there, since we need that next. Tell me the result of both, and your OS if you haven't yet.

yep next

Good, git's sorted. Next up is the fork and clone.

Step 2: Fork the repo

1. Log into GitHub with your own account (SuhanaAGhosh).
2. Open github.com/SETAPESU26/53_arm_wrestle
3. Click Fork (top right), then Create fork. Leave the defaults.
4. Check that the URL now reads `github.com/SuhanaAGhosh/53_arm_wrestle`. If it still says SETAPESU26, the fork didn't happen.

Step 3: Clone it

In your terminal (a fresh one, after the git install):

```
cd Desktop
git clone https://github.com/SuhanaAGhosh/53_arm_wrestle.git

cd 53_arm_wrestle
```

If a browser window pops up asking you to sign in to GitHub, do that. Then run `dir` (Windows) or `ls` (Mac). You should see `game`, `main.py` and `README.md`.

Step 4: Check Python and install pygame

```
python --version

pip install pygame
```

You need Python 3.10 or higher. If `python` isn't recognized, try `python3`, and use `pip3` instead of `pip`.

Step 5: Run it

```
python main.py
```

Click the window and mash Left/Right alternately. If the bug is there, the hands slide toward the computer's side.

Tell me where you got to, and paste any errors exactly as they appear. If the game runs, we go straight to the before video.

im using the github id ryonahh

Got it, I'll use ryonahh from here on. Same steps, corrected:

Step 2: Fork

1. Log into GitHub as ryonahh.
2. Open github.com/SETAPESU26/53_arm_wrestle, click Fork, then Create fork.
3. The URL should now read github.com/ryonahh/53_arm_wrestle. If it still says SETAPESU26, it didn't fork.

Step 3: Clone

```
cd Desktop
git clone https://github.com/ryonahh/53_arm_wrestle.git

cd 53_arm_wrestle
```

Sign in via the browser popup if it asks. `dir` (Windows) or `ls` (Mac) should show `game`, `main.py` and `README.md`.

Step 4: Python and pygame

```
python --version

pip install pygame
```

You need Python 3.10 or higher. Try `python3` and `pip3` if the plain ones aren't recognized.

Step 5: Run

```
python main.py
```

Click the window and mash Left/Right. The hands should slide toward the computer's side, which is the bug.

Also, when we get to git config later, use the email tied to ryonahh, not any other account. Otherwise the commits won't be attributed to that profile.

Tell me where you get stuck, and paste any errors exactly.

Images hidden when shared

this is what i hv till now

Fork is done and it's under your account. Two things I need to flag, though. The repo name has your SRN in it, so the clone URL is different. And it shows 3 commits (last one "Update game_engine.py", last week), while the copy I read earlier showed 1. I may have analyzed a stale version, so I'm re-checking your fork's code before we rely on it.

Read 2 pages

Read 2 pages

Good thing I checked. The README I read earlier was an old version, and I gave you wrong info. The code is the same, but the tasks changed:

Task	Actual requirement
1	Fix the inverted push (same as before)
2	AI surge cycle: periodic high-power surge, then a cooldown with reduced resistance

-
- 3 Warning indicators: show when the AI surges, and show an exhaustion state when player stamina is too low to push
-

- 4 Counter-surge bonus: if you push right as the AI's surge ends and cooldown begins, you get a stamina recovery boost and double push strength temporarily

Tasks 3 and 4 are no longer duplicates, and I was wrong to tell you to merge them. Your plan is now 4 tasks, 4 separate commits. They build on each other (3 needs 2's surge state, 4 needs 3's states), so we do them in order.

Your clone URL is different because the repo name has your SRN in it:

```
cd Desktop
git clone https://github.com/ryonahh/PES1UG24AM233_53_arm_wrestle.git
```

```
cd PES1UG24AM233_53_arm_wrestle
```

Then:

```
pip install pygame
```

```
python main.py
```

My earlier tuning concern still holds, and it matters more now. Stamina regenerates about 48/sec and only drains about 2 per press, so you'll basically never hit the exhaustion state. We'll rebalance it in Task 3, otherwise the "after" video can't show it.

Have you cloned yet? Tell me if the game opened, or paste any error.

OKAY the game is working after i did `python main.py` and i got a window, but now the codes and game in turn has many bugs and my task is to fix that

Good, game's running. Before we fix anything, we need the "before" video, and it has to be recorded before any code changes.

Record the "before" video (10 sec)

1. Start the screen recorder: Win+G (Game Bar), or Win+Alt+R to start/stop recording directly.
2. Run `python main.py`, click the game window, and mash Left/Right alternately for about 10 seconds.
3. What the video must show: the hands sliding toward the computer's side while you're pushing, ending in "COMPUTER WINS!". Press R once if you want to show the rematch works.
4. Stop recording. Game Bar saves to `Videos\Captures`. Rename the file to `before.mp4`.

Don't touch the code until this is saved.

While you wait on it: tell me what you saw

You said there are "many bugs." The README only claims one deliberate bug, plus features to add. Before we start editing, tell me what you noticed beyond the hands moving the wrong way. For example:

- Does the stamina bar ever visibly drop?
- Does the game feel too easy or too hard?
- Anything that crashes or looks off?

If it's only the reversed push, the "many bugs" are really the missing features (Tasks 2-4), and that's fine. I'd rather know what you actually see than assume.

Once `before.mp4` is saved, tell me and we start Task 1. For the lab's "3-4 prompts per task" rule, your first prompt should look like this:

"In game/game_engine.py, handle_event adds +4.2 to arm_position for both Left and Right keys. Player wins at arm_position <= -100, so pressing should subtract 4.2. Fix it and show only the changed lines."

That's specific enough to get it done in one shot.

okay lets go ahead with task one, explain what i should do to implement task 1 in simple terms

Quick check first: is `before.mp4` saved? If not, do that now, because once you edit the code the bug is gone and you can't re-record it.

What's wrong, in plain terms

In this game, the player's side is the left and a win means `arm_position` reaches -100. So each key press should push the number down (negative). The code pushes it up by adding 4.2, which helps the computer win.

The fix

Open `game/game_engine.py` (VS Code, Notepad, whatever you use). Find `handle_event`. There are two identical lines, one under the Left key and one under the Right key:

```
python
```

```
self.arm_position += 4.2
```

Change both to:

```
python
```

```
self.arm_position -= 4.2
```

That's the entire fix: two characters changed (+ to -), in two places. Don't touch anything else. Save the file.

Test it

```
python main.py
```

Mash Left/Right alternately. The hands should now slide toward the left (your side), and you should reach "PLAYER WINS THE MATCH!" in a few seconds. Press R to confirm the rematch still works.

If it works, commit it:

```
git add game/game_engine.py
git commit -m "Task 1: fix inverted arm push bug"

git push
```

If `git commit` complains about "Please tell me who you are", run these once, using the email tied to ryonahh, then redo the commit:

```
git config --global user.name "ryonahh"

git config --global user.email "your-ryonahh-email@example.com"
```

One honest note

After this fix the game gets very easy. You'll win in about 4 seconds, which is why Task 2 (AI surge) matters. Don't worry about balance yet. Just confirm the direction is right.

Tell me when it's committed, or what you saw if it didn't behave. Then we do Task 2.

Images hidden when shared

can you guide me from here

You're in a GitHub Codespace (browser VS Code), which changes two things.

1. You've got a duplicate. You ran `git clone` *inside* a repo that's already cloned, so there's a nested `PES1UG24AM233_53_arm_wrestle` folder in your Explorer. Delete it so it doesn't confuse things:

```
rm -rf PES1UG24AM233_53_arm_wrestle
```

Run that from the terminal you already have open (you're at the repo root). The folder is a fresh copy with no changes, so nothing's lost.

2. Codespaces can't show the game window. It's a cloud Linux box with no screen, so `python main.py` won't open anything there. Your window earlier must have come from your laptop. So the split is:

- Codespace: edit the code and commit/push.
- Your laptop: run the game, record the videos.

Do Task 1 now, in the Codespace

1. Click `game/game_engine.py` in the Explorer.
2. Press Ctrl+H (find and replace). Find `self.arm_position += 4.2`, replace with `self.arm_position -= 4.2`, then click Replace All. It should say 2 replacements. If it says anything else, stop and tell me.
3. Ctrl+S to save.
4. In the terminal:

```
git add game/game_engine.py
git commit -m "Task 1: fix inverted arm push bug"
```

```
git push
```

Codespaces sets up your git login automatically, so you shouldn't hit the identity error.

Then test on your laptop

In your local project folder (the one where the game opened):

```
git pull
```

```
python main.py
```

Mash Left/Right. The hands should now go toward the left (your side) and you win in a few seconds.

Did you record `before.mp4` yet? Your laptop copy still has the bug until you `git pull`, so you have time. Record it first, then pull.

Tell me once the push goes through, or paste any error.

i have used the git pull, but the computer still wins, is that alright? if yes lets continue with task 2

Not alright. Don't move to Task 2 yet. After the fix, mashing should win in about 4 seconds, so something didn't land. Run these in your laptop terminal, in the project folder:

```
git pull
```

```
git log --oneline -3
```

```
findstr "arm_position" game\game_engine.py
```

(On Mac/Linux, use `grep "arm_position" game/game_engine.py` for the last one.)

What you should see

- `git log` should list “Task 1: fix inverted arm push bug” at the top.
- The `findstr` output should show two lines with `== 4.2.`

What each result means

- Still += 4.2, and no Task 1 commit in the log: the push from the Codespace never went through, or it didn't commit. Go back to the Codespace, run `git status` and `git log --oneline -3` there, and paste me the output.
- `pull` says "Already up to date" but the file still has +=: same cause. The change never reached GitHub.
- Shows -= 4.2 in both places: the code is fixed and it's a play issue. You must alternate Left, Right, Left, Right. Pressing the same key twice in a row does nothing. Aim for 5+ alternating presses per second. If you're slower than about 3 presses/sec, the AI still wins.

Also confirm you restarted the game after pulling. An already-open window keeps running the old code.

Paste the output of those three commands and I'll tell you exactly which case it is.

```
C:\Users\ryona\Desktop\PES1UG24AM233_53_arm_wrestle>git log --oneline -3
3c93a74 (HEAD -> main, origin/main, origin/HEAD) Task 1: fix inverted arm push bug
de3364b Update game_engine.py
ca2c291 Update README.md
```

```
C:\Users\ryona\Desktop\PES1UG24AM233_53_arm_wrestle>ndstr "arm_position"
game\game_engine.py
'ndstr' is not recognized as an internal or external command,
operable program or batch file.
this is what my cmd told me
```

Good news: the commit did land. Your log shows "Task 1: fix inverted arm push bug" at the top, and your local copy matches GitHub. So the push and pull worked.

The `findstr` just got typed as `ndstr` (missing the "f"). Run it again:

```
findstr "arm_position" game\game_engine.py
```

You want to see two lines with -= 4.2.

If it shows `-= 4.2` on both

The code is fixed and the problem is how you're playing or running it:

1. Close the old game window completely. A window opened before the pull keeps running the old code.
2. Run `python main.py` fresh.
3. Alternate Left, Right, Left, Right. Pressing the same key twice does nothing. You need roughly 5+ alternating presses per second.

Watch the hands. If they now creep toward the left (your side), even slowly, the fix works and you were just too slow. The AI still pushes the other way all the time, so mashing at a lazy pace loses.

If it still shows `+= 4.2`

The Replace All didn't do what we expected despite the commit. Paste the output and I'll find out why.

Paste the `findstr` result, and tell me which way the hands moved on the fresh run.

there is something else at play because none of my keys are letting me win even though it is `-=4.2`

Let me check what's actually on GitHub rather than guess.

Read a page

Read a page

That shows `+= 4.2`, but this URL was fetched earlier and may be cached. I'll check the commit's actual diff instead.

Read a page

Read a page

The code is correct. The commit diff on GitHub shows both lines changed to `-- 4.2`, and the earlier fetch I did was a stale cached copy. So the fix is in, and the problem is the game's timing, not Task 1.

What's happening

The AI pushes from the moment the window opens, at about 13.6 units/sec. With no input, the computer wins in about 7 seconds. If you click into the window and start mashing late, it may already be over. Keys do nothing on the game-over screen except R.

Each alternating press only moves you 4.2, so you need more than 3.3 presses/sec just to break even. At a normal mash, about 5/sec, you only gain about 7 units/sec net, which means roughly 14 seconds to win. And the AI has a head start. I said earlier you'd win in 4 seconds, but that assumed about 10 presses/sec, so I overstated it.

Isolate it: remove the AI for one test

1. In `game/game_engine.py`, change `self.ai_strength = 0.35` to `self.ai_strength = 0.0`. Save it.
2. Run `python main.py`, click the window first, and alternate Left/Right slowly.
3. The hands should now slide left steadily and you should get "PLAYER WINS THE MATCH!"

If that works, the fix is fine and it was just speed and timing. Then undo the test change:

```
git checkout game/game_engine.py
```

If the hands don't move left even with the AI off, tell me exactly what you see. That would mean the window isn't receiving your keys, and I'd look at focus and your Python environment next.

Also try this

Press R to rematch right before you start mashing. That gives you a fresh 7-second window instead of whatever's left of the first game.

Tell me the result of the AI-off test.

am i supposed to use my mouse or keyboard

Keyboard. Use the Left Arrow and Right Arrow keys, alternating: Left, Right, Left, Right. The mouse does nothing in this game.

You only need the mouse once, to click the game window so it has focus. After that, hands on the keyboard.

- Pressing the same arrow twice in a row does nothing, so you have to alternate.
- Holding a key down doesn't count either. Every press has to be a fresh tap.
- R restarts after a game ends.

If you were using the mouse this whole time, that explains everything. Try the AI-off test now, tapping the arrows, and tell me what happens.

working! do i need a recording

Not for Task 1. The handout needs only two videos total: one before (the bug) and one after (everything working), recorded once at the very end.

But you still need the before video, and your laptop copy now has the fix. So you have to temporarily roll the file back to record the bug. Here's how, safely:

1. Undo the AI-off test (if you left it on)

```
git checkout -- game/game_engine.py
```

This resets the file to the committed, fixed version.

2. Roll back to the buggy version to record

```
git checkout de3364b -- game/game_engine.py
```

```
python main.py
```

- Start recording (Win+Alt+R) and click the game window.
- Alternate Left/Right arrow keys for about 10 seconds.
- The hands should slide toward the computer's side until "COMPUTER WINS!" appears. That's the bug.
- Stop recording and rename the file to `before.mp4`.

3. Restore the fixed version

```
git checkout main -- game/game_engine.py
```

```
git status
```

`git status` should say nothing to commit, working tree clean. If it lists `game_engine.py` as modified, tell me before doing anything else.

Once that's done and `before.mp4` is saved somewhere safe, we start Task 2 (AI surge). Tell me when you're clean.

i have the before video ill take recording of this too and then we start with task 2

Go for it. Extra clips are harmless, but the handout only needs before and after, so don't stress about more.

Before you move on, check you're clean:

```
git status
```

It should say nothing to commit, working tree clean. If `game_engine.py` shows as modified, that's the leftover AI-off test (`ai_strength = 0.0`) or the rollback. Run `git checkout main -- game/game_engine.py` to reset it. That setting must never get committed.

Task 2 plan

The AI gets a three-phase cycle, tracked in the engine:

- Building: normal pressure while its energy meter fills (about 4 sec)
- Surge: about 1.5 sec at roughly 4x force
- Cooldown: about 2 sec at reduced force, then back to building

`reset()` also needs to restart the cycle, or rematches begin mid-surge. I'll keep the phase as a named state ("BUILDING", "SURGE", "COOLDOWN") because Tasks 3 and 4 hang off it: the warning indicator reads it, and the counter-surge bonus fires at the surge-to-cooldown switch.

Tell me when you're clean and I'll write the code. I'll show only the changed parts so you can review them before pasting. For your chat record, this is the prompt that matches what we're doing:

“Task 2: in game/game_engine.py, replace the constant AI force in update() with a surge cycle: build up, 1-2 sec surge at higher force, then cooldown at reduced force. Track the phase as a state variable and reset it in reset().”

im clean lets start step by step in details and give me cases for each step if i dont have something done, or catch an error

Here's Task 2, one step at a time. Each step has a "stuck?" line.

Step 0: Work on your laptop from now on

The Codespace can't show the game window, so editing there means commit, push, pull, then test. Skip all that. Edit, run and commit on the laptop.

In your project folder:

```
code game\game_engine.py
```

- "code is not recognized": VS Code isn't installed. Use notepad game\game_engine.py. It works, but be careful with indentation (4 spaces, never Tab).
- Push asks you to sign in later: do it through the browser popup.

Step 1: Add the surge settings to `__init__`

Find this line near the top:

```
python
```

```
self.ai_strength = 0.35
```

Click at the end of that line, press Enter, and paste this below it, at the same indent:

```
python
```

```
    # --- AI surge cycle (Task 2) ---
    self.build_ms = 4000
    self.surge_ms = 1500
    self.cooldown_ms = 2000
    self.phase_multipliers = {"BUILDING": 1.0, "SURGE": 3.5,
"COOLDOWN": 0.3}
    self.ai_phase = "BUILDING"

    self.phase_start = pygame.time.get_ticks()
```

This sets up three phases and how hard the AI pushes in each. The times are in milliseconds.

- **Stuck?** `IndentationError` later means your pasted lines don't line up with `self.ai_strength`. **Select them and fix the spaces.** `AttributeError: no attribute 'ai_phase'` means this step is missing, or you pasted it outside `__init__`, so it has to sit above `def handle_event`.

Step 2: Add two new methods

Find `def update(self):` and paste this directly above it, with a blank line before and after. `def` should line up with the other `defs` (4 spaces).

python

```
def set_ai_phase(self, phase):
    self.ai_phase = phase
    self.phase_start = pygame.time.get_ticks()
    print("AI phase:", phase) # TEMP: delete after testing

def update_ai_phase(self):
    elapsed = pygame.time.get_ticks() - self.phase_start
    if self.ai_phase == "BUILDING" and elapsed >= self.build_ms:
        self.set_ai_phase("SURGE")
    elif self.ai_phase == "SURGE" and elapsed >= self.surge_ms:
        self.set_ai_phase("COOLDOWN")
    elif self.ai_phase == "COOLDOWN" and elapsed >= self.cooldown_ms:
        self.set_ai_phase("BUILDING")
```

`update_ai_phase` checks how long the current phase has run and moves to the next one: BUILDING, then SURGE, then COOLDOWN, then back to BUILDING.

- **Stuck?** `IndentationError: unexpected indent` means the `def` lines aren't at 4 spaces. A “method not found” error later means you pasted this inside another method.

Step 3: Use the phase in `update()`

Inside `update`, find these two lines:

python

```
ai_variance = random.uniform(0.3, 1.0)

self.arm_position += self.ai_strength * ai_variance
```

Replace them with:

python

```
self.update_ai_phase()
ai_variance = random.uniform(0.3, 1.0)
force = self.ai_strength * ai_variance *
self.phase_multipliers[self.ai_phase]

self.arm_position += force
```

- **Stuck?** `KeyError: 'SURGE'` (or similar) means a phase name is mistyped somewhere. Names must match exactly, in capitals. If the file has leftover `ai_variance` lines duplicated, delete the extras.

Step 4: Reset the cycle in `reset()`

In `def reset`, add one line at the end, after `self.game_state = "PLAYING"`:

python

```
self.set_ai_phase("BUILDING")
```

`self.game_state = "PLAYING"` also appears in `__init__`. Make sure you're in `reset`, the shorter one near the bottom of the logic, above `def render`. Without this, a rematch can start mid-surge.

Step 5: Save and test

Ctrl+S, then:

python main.py

Look at the terminal (not the game window) while you mash. You should see, with these timings:

```
AI phase: SURGE          (about 4 s in)
AI phase: COOLDOWN       (1.5 s later)
AI phase: BUILDING       (2 s later)
```

```
AI phase: SURGE          (and so on)
```

In the game, the hands should lurch hard toward the computer during SURGE, and your pushing should feel much easier during COOLDOWN.

What you see	Cause
Traceback in the terminal	Paste it to me exactly, the last 5 lines matter most
Window opens, no prints at all	Not saved, <code>update_ai_phase()</code> call missing from Step 3, or an old window still open
Prints but you can't feel anything	Mash and watch the hands. SURGE is about 3.5x harder
Stuck on BUILDING forever	Step 2's <code>if</code> conditions mistyped
Game ends before a surge	The game is over after about 7 s with no input, so press R and start mashing right away

Step 6: Clean up and commit

1. Delete the `print("AI phase:", phase) # TEMP... line`, then save.
2. Quick re-run to make sure it still works (no prints now).
3. Commit:

```
git add game/game_engine.py
git commit -m "Task 2: add AI surge/cooldown cycle"

git push
```

Then `git status` should say working tree clean.

Tell me where you are after Step 5, or paste any error. Don't start Task 3 until this is committed, since it builds directly on the phase state.

Images hidden when shared

i didnt see anything on the terminal and this is the code i put in my github codespace

The part I can see (lines 54-66) is correct. The likely reason you saw nothing is that you edited in the Codespace, but you ran the game on your laptop. Your laptop still has the old file, since nothing was pushed. The Codespace has no screen, so the game can't run there. (I'd said to edit on the laptop to avoid this, but this is fixable.)

The tab also shows "1, M": one problem flagged, file modified. I can't see Steps 1, 3 or 4 in your screenshot, so let's check them before testing.

Step A: Verify the whole edit, in the Codespace terminal

```
python -m py_compile game/game_engine.py

grep -n "phase" game/game_engine.py
```

- First command prints nothing: syntax is fine.
- First command prints an error: paste it to me, that's the "1 problem."
- `grep` output: paste all of it. I can tell from it which steps are in the file and which are missing.

Also open the Problems panel (Ctrl+Shift+M) and tell me what the 1 problem says.

Step B: Once Step A looks right, get it onto your laptop

In the Codespace terminal:

```
git add game/game_engine.py
git commit -m "Task 2: add AI surge/cooldown cycle"

git push
```

Then on your laptop:

```
git pull

python main.py
```

Watch the laptop terminal. You should see `AI phase: SURGE` about 4 seconds in, then `COOLDOWN`, then `BUILDING`.

If testing shows something broken, fix it and make another commit, like “Task 2: fix surge timing”. Extra commits per task are fine, since the handout just wants each task in its own commit(s).

Step C: If the laptop prints nothing after the pull

- Run `git log --oneline -2` and check that your Task 2 commit is on top. If it isn't, the push or pull didn't happen.
- Close any old game window and rerun.

Paste the Step A output and I'll tell you if anything's missing.

Import "pygame" could not be resolved is what the problems panel says

That's harmless. It's an editor warning, not a bug in your code. pygame isn't installed in the Codespace (only on your laptop), so VS Code can't resolve the import. The game will run fine on your laptop.

If you want the warning gone, run this in the Codespace terminal:

```
pip install pygame
```

That's optional. The Codespace still can't open the game window.

What I still need

That warning isn't the same as a syntax error, so run these in the Codespace terminal:

```
python -m py_compile game/game_engine.py
```

```
grep -n "phase" game/game_engine.py
```

- `py_compile` printing nothing means the syntax is fine.
- Paste the `grep` output. I can't see Steps 1, 3 or 4 in your screenshot, and this shows me whether they're in the file:
 - Step 1: `ai_phase, phase_multipliers` etc. in `__init__`
 - Step 3: `self.update_ai_phase()` and `phase_multipliers[self.ai_phase]` inside `update`
 - Step 4: `self.set_ai_phase("BUILDING")` in `reset`

If everything checks out, go to Step B from before: commit and push from the Codespace, then `git pull` and `python main.py` on your laptop.

Paste the `grep` output and I'll confirm it's complete.

i have seen surge, cooldown, building now remind and guide me through the next phase

Nice, the cycle works. Two parts: wrap up Task 2 properly, then Task 3.

Finish Task 2

Step 1: Move to the laptop for editing

You've been editing in the Codespace and testing on the laptop, which forces a push before every test. From here, edit on the laptop. Open a terminal in your project folder:

```
git pull
```

```
git log --oneline -3
```

- Top commit is Task 2: ...: already committed. The cleanup below is a small second commit, which is fine.
- No Task 2 commit in the log: you haven't pushed it. Tell me and we'll sort it before anything else.

Step 2: Delete the debug print

```
code game\game_engine.py
```

Delete the `print("AI phase:", phase) # TEMP... line in set_ai_phase`. Save with Ctrl+S.

- **code not recognized:** use `notepad game\game_engine.py`.

Step 3: Test the feel

`python main.py`, click the window, and mash for about 20 seconds. You want to see:

- Hands lurch toward the computer for ~1.5 s every ~7.5 s (the surge)
- Easy pushing for ~2 s after each surge (the cooldown)
- No terminal prints

Nothing visible means a stale window is still open. Close it and rerun.

Step 4: Commit

```
git add game/game_engine.py
```

```
git commit -m "Task 2: remove debug print"
```

```
git push
```

Then `git status` should say clean.

Task 3: Warnings and exhaustion

What it adds: a flashing “AI SURGE” banner during surges, plus an “EXHAUSTED!” label and flashing red bar when you’re out of stamina.

Why it also touches stamina: right now regen (48/sec) beats drain, so you can never get exhausted and the indicator would never show in your video. So we rebalance, and add a recovery threshold so the lockout lasts about 2 seconds instead of one frame.

All edits are in `game/game_engine.py`.

Step 1: New variables in `__init__`

Paste directly under the `self.phase_start = ...` line, same indent:

python

```
self.exhausted = False
self.exhaust_below = 10

self.recover_at = 30
```

- **Stuck?** `AttributeError: exhausted` later means this step is missing or outside `__init__`.

Step 2: Make the drain bigger

Press `Ctrl+H`. Find `self.stamina - 2.0`, replace with `self.stamina - 3.0`, click `Replace All`. It should say 2 replacements. Any other number, stop and tell me.

Step 3: Block input while exhausted

In `handle_event`, find:

python

```
if self.stamina <= 10:  
    return
```

Replace with:

python

```
if self.exhausted or self.stamina <= self.exhaust_below:  
    return
```

Step 4: Slower regen plus the exhausted flag

In `update`, find:

python

```
if self.stamina < self.max_stamina:  
    self.stamina = min(self.max_stamina, self.stamina + 0.8)
```

Replace with:

python

```
if self.stamina < self.max_stamina:  
    self.stamina = min(self.max_stamina, self.stamina + 0.2)  
  
if self.stamina <= self.exhaust_below:  
    self.exhausted = True  
elif self.exhausted and self.stamina >= self.recover_at:  
    self.exhausted = False
```

Once exhausted, you stay locked out until stamina climbs back to 30, which takes about 2 s.

Step 5: Reset it

In `reset()`, add at the end, under `self.set_ai_phase("BUILDING")`:

python

```
self.exhausted = False
```

Step 6: Draw the indicators in `render()`

6a. Find:

python

```
bar_color = (60, 210, 100) if self.stamina > 25 else (220, 60, 60)
```

Replace with:

python

```
blink = (pygame.time.get_ticks() // 200) % 2 == 0
if self.exhausted:
    bar_color = (255, 40, 40) if blink else (120, 20, 20)
else:

    bar_color = (60, 210, 100) if self.stamina > 25 else (220, 60,
60)
```

6b. Find this line (right below it):

python

```
pygame.draw.rect(screen, bar_color, stamina_fill, border_radius=6)
```

Paste this below it, before `if self.game_state == "GAME_OVER":`:

python

```
if self.exhausted:
    ex_surf = self.font_med.render("EXHAUSTED!", True, (255, 80,
80))
    screen.blit(ex_surf, (400, 450))

if self.ai_phase == "SURGE" and self.game_state == "PLAYING" and
blink:
    warn = self.font_big.render("!! AI SURGE !!", True, (255, 170,
40))
    screen.blit(warn, (self.width // 2 - warn.get_width() // 2,
60))
```

```
pygame.draw.rect(screen, (255, 120, 40), pygame.Rect(0, 0,
self.width, self.height), width=6)
```

- Stuck? `NameError: blink` means 6a is missing, or 6b is placed above it.

Step 7: Test

Save, then `python main.py`.

You do	You should see
Wait ~4 s	Flashing orange “!! AI SURGE !!” plus an orange border for 1.5 s
Mash hard (8+ presses/sec) for ~6 s	Bar turns red and flashes, “EXHAUSTED!” appears, your keys stop working
Stop pressing for ~2 s	Warning clears and keys work again

Problem	Cause
Never gets exhausted	Mash faster, you need 8+ presses/sec. Also check Step 2 said 2 replacements
Game ends before you get exhausted	Press R and mash right away. Cooldown is the best time to push
Keys stay dead	Step 3 or 4 mistyped, or <code>recover_at</code> isn't 30
Traceback	Paste the last 5 lines to me

Step 8: Commit

```
git add game/game_engine.py
git commit -m "Task 3: add surge warning and exhaustion indicator"

git push
```

Tell me how the test goes. Task 4 builds on the SURGE-to-COOLDOWN switch from Task 2, so don't start it until this is committed.

Your branch is up to date with 'origin/main'.

Untracked files:

(use "git add <file>..." to include in what will be committed)

game/__pycache__/

nothing added to commit but untracked files present (use "git add" to track)

wait i got this after git status

That's fine. `game/__pycache__/` is a cache folder Python generates automatically when you run the game. It's not your code, so don't commit it.

- Always add files by name (`git add game/game_engine.py`), **never** `git add ..`. Then the cache folder can't sneak in.
- If you'd rather the warning go away, create a file named `.gitignore` containing `__pycache__/,` and commit it separately: `git add .gitignore, then git commit -m "Add .gitignore".` That's optional.

What it also tells me

"Nothing added to commit" means no tracked file has unsaved changes. So either you've already removed the debug print and committed it, or you haven't touched it yet. Check:

```
grep -n "print" game/game_engine.py
```

(On Windows cmd: `findstr "print" game\game_engine.py`)

- Nothing printed: the debug print is gone and Task 2 is done. Go straight to Task 3, Step 1.

- `print("AI phase:" ... still shows: delete that line, save, then:`

```
git add game/game_engine.py
```

```
git commit -m "Task 2: remove debug print"
```

```
git push
```

Tell me which one you got, and then start Task 3 from Step 1.

also from now on specify where i have to run a code, github codespace or cmd okay?

Done. I'll tag every command from here on:

- [CMD] = Windows Command Prompt on your laptop, in
C:\Users\ryona\Desktop\PES1UG24AM233_53_arm_wrestle
- [CODESPACE] = the terminal in your browser

From now on, do everything in [CMD] on your laptop, including editing. That's where the game runs, so you skip the commit, push and pull cycle between every test. The Codespace is only needed if something's wrong with GitHub itself.

Your next step

Check if the debug print is still in the file:

[CMD]

```
findstr "print" game\game_engine.py
```

- Nothing printed: Task 2 is done. Start Task 3, Step 1.
- `print("AI phase:" ... shows up: open the file with code
game\game_engine.py, delete that line, save, then:`

[CMD]

```
git add game/game_engine.py
git commit -m "Task 2: remove debug print"
```

```
git push
```

One more setup check: your laptop copy has to contain the Task 2 code, because you wrote it in the Codespace. Confirm it with:

[CMD]

```
git pull
```

```
git log --oneline -3
```

You should see a “Task 2” commit near the top. If you don’t, tell me before you edit anything.

Which result did you get on the `findstr`?

u meant to put that in github codespace right? anyway

Already up to date.

fa53cc1 (HEAD -> main, origin/main, origin/HEAD) Task 2: remove debug print

7ea2f80 Task 2: add AI surge/cooldown cycle

3c93a74 Task 1: fix inverted arm push bug

is what i got

No, that one was [CMD]. `findstr` is a Windows command and wouldn’t work in the Codespace. The `git pull` and `git log` you ran in CMD are what mattered anyway, and the output is exactly right:

- Task 1 
- Task 2  (cycle added, debug print removed)
- Laptop and GitHub are in sync

Now Task 3. I'll tag every step: [CMD] for Command Prompt on your laptop, [EDITOR] for edits inside `game_engine.py`.

Step 0: Open the file

[CMD]

```
code game\game_engine.py
```

If code isn't recognized, use `notepad game\game_engine.py`.

Step 1: New variables in `__init__`

[EDITOR] Find the line `self.phase_start = pygame.time.get_ticks()` in `__init__`. Paste directly under it, same indent:

```
python
    self.exhausted = False
    self.exhaust_below = 10

    self.recover_at = 30
```

Step 2: Bigger stamina drain

[EDITOR] Press Ctrl+H. Find `self.stamina - 2.0`, replace with `self.stamina - 3.0`, click Replace All. It must say 2 replacements. Any other number, stop and tell me.

Step 3: Block input while exhausted

[EDITOR] In `handle_event`, find:

```
python
    if self.stamina <= 10:

        return
```

Replace with:

python

```
if self.exhausted or self.stamina <= self.exhaust_below:  
  
    return
```

Step 4: Slower regen plus exhausted flag

[EDITOR] In `update`, find:

python

```
if self.stamina < self.max_stamina:  
  
    self.stamina = min(self.max_stamina, self.stamina + 0.8)
```

Replace with:

python

```
if self.stamina < self.max_stamina:  
    self.stamina = min(self.max_stamina, self.stamina + 0.2)  
  
if self.stamina <= self.exhaust_below:  
    self.exhausted = True  
elif self.exhausted and self.stamina >= self.recover_at:  
  
    self.exhausted = False
```

Step 5: Reset it

[EDITOR] In `reset()`, add at the end, under `self.set_ai_phase("BUILDING")`:

python

```
self.exhausted = False
```

Step 6: Indicators in `render()`

[EDITOR] 6a. Find:

python

```
bar_color = (60, 210, 100) if self.stamina > 25 else (220, 60, 60)
```

Replace with:

python

```
        blink = (pygame.time.get_ticks() // 200) % 2 == 0
        if self.exhausted:
            bar_color = (255, 40, 40) if blink else (120, 20, 20)
        else:

            bar_color = (60, 210, 100) if self.stamina > 25 else (220, 60,
60)
```

[EDITOR] 6b. Find `pygame.draw.rect(screen, bar_color, stamina_fill,`
`border_radius=6)`. Paste below it, before `if self.game_state == "GAME_OVER"::`

python

```
        if self.exhausted:
            ex_surf = self.font_med.render("EXHAUSTED!", True, (255, 80,
80))
            screen.blit(ex_surf, (400, 450))

        if self.ai_phase == "SURGE" and self.game_state == "PLAYING" and
blink:
            warn = self.font_big.render("!! AI SURGE !!", True, (255, 170,
40))
            screen.blit(warn, (self.width // 2 - warn.get_width() // 2,
60))

            pygame.draw.rect(screen, (255, 120, 40), pygame.Rect(0, 0,
self.width, self.height), width=6)
```

Save with Ctrl+S.

Step 7: Test

[CMD]

python main.py

You do

You should see

Wait ~4 s	Flashing orange “!! AI SURGE !!” plus an orange border for 1.5 s
Mash 8+ presses/sec for ~6 s	Flashing red bar, “EXHAUSTED!” label, and your keys stop working
Stop pressing ~2 s	Warning clears and keys work again
Problem	Cause
Never gets exhausted	Mash faster, and check Step 2 said 2 replacements
<code>NameError: blink</code>	6a missing, or 6b placed above it
Keys stay dead	Step 3 or 4 mistyped
Traceback	Paste the last 5 lines to me

Step 8: Commit

[CMD]

```
git add game/game_engine.py
git commit -m "Task 3: add surge warning and exhaustion indicator"

git push
```

Tell me how the test goes, and paste any error exactly.

i dont see my bars changing colours tho

Most likely your edits are fine and you never reach the exhausted state. That's partly my mistake: my test numbers were off.

At a decent mash, about 8 presses/sec, you push the arm left so fast that you win in about 5 seconds, before stamina has dropped enough to turn the bar red. Drain was too weak for how strong your presses are.

Step 1: Confirm the edits are in

[CMD]

```
findstr "exhausted" game\game_engine.py
```

```
findstr "stamina - " game\game_engine.py
```

- **First command:** you should see about 8 lines (`self.exhausted = False, if self.exhausted or ..., bar_color, the EXHAUSTED! render, and so on`). Nothing printed means Steps 1-6 didn't save. Reopen the file and check.
- **Second command:** you should see `self.stamina - 3.0` twice. If it still says `2.0`, Step 2 didn't apply.

Step 2: Fix the balance (this one stays)

[EDITOR] In `game_engine.py`, press Ctrl+H. Find `self.stamina - 3.0`, replace with `self.stamina - 5.0`, click Replace All. It should say 2 replacements. Save with Ctrl+S.

New numbers: drain is 5 per press and regen is 12/sec, so at 8 presses/sec you exhaust in about 3 seconds, mid-fight, which is what the video needs.

Step 3: Test

[CMD]

```
python main.py
```

Click the window, press R for a fresh game, then mash alternating Left/Right as fast as you can.

You do	You should see
Mash fast for ~3 s	Bar turns red and flashes, “EXHAUSTED!” appears, keys die
Stop ~2 s	Warning clears, keys work
Wait ~4 s into a game	Flashing orange “!! AI SURGE !!” banner and border

Quick test shortcut (optional)

If you want to see the red bar immediately, [EDITOR] temporarily set `self.stamina = 100.0` in `__init__` to `self.stamina = 15.0`. One press should trigger exhaustion.

Undo it right after:

[CMD]

```
git checkout game/game_engine.py
```

Don’t do that if you haven’t committed Task 3 yet, because it would wipe your edits. Just change the number back by hand.

Tell me what the `findstr` commands showed and what you saw in the game. If the surge banner never appeared either, say so, because that means a different step is missing.

Images hidden when shared

see this is the output in cmd

That screenshot answers it, and it's not a balance problem. I guessed wrong earlier.

`findstr "exhausted"` printed nothing, so none of the Task 3 code is in the file on your laptop. The rest of the output confirms it: `drain` is still `- 2.0`, `regen` is still `+ 0.8`, and the input check is still `<= 10`. The bar can't change colour because the code that does it isn't there.

(The second `findstr` was noisy because of my pattern. Spaces make `findstr` search for either word. Use `findstr /C:"stamina - 5.0" game\game_engine.py` for exact matches.)

Where did you make the Task 3 edits?

Case A: in the Codespace. The edits are there but never reached your laptop.

[CODESPACE]

```
git status
```

If `game/game_engine.py` shows as modified:

```
git add game/game_engine.py
git commit -m "Task 3: add surge warning and exhaustion indicator"
```

```
git push
```

[CMD]

```
git pull
```

```
findstr "exhausted" game\game_engine.py
```

You should now see about 8 lines. Then fix the balance in the next section.

Case B: on the laptop, but never saved (or never made). Redo Task 3 Steps 0-6 from my earlier message, [CMD] to open the file and [EDITOR] for the edits. Press `Ctrl+S` at the

end. Make one change: in Step 2, replace `self.stamina - 2.0` with `self.stamina - 5.0` (not 3.0). It must say 2 replacements.

Balance (needed in both cases)

The drain has to be 5.0 per press for you to get exhausted mid-fight.

[EDITOR] Ctrl+H, find `self.stamina - 3.0`, replace with `self.stamina - 5.0`. Skip this if you took Case B.

Verify before testing

[CMD]

```
findstr /C:"exhausted" game\game_engine.py  
findstr /C:"stamina - 5.0" game\game_engine.py
```

- First: about 8 lines.
- Second: exactly 2 lines.

Then `python main.py`, press R, and mash fast. You should get a flashing red bar and “EXHAUSTED!” after about 3 seconds.

Also, the AI phase: SURGE prints at the top of your screenshot should be from before you removed the debug print. If a fresh run still prints them, tell me.

Tell me which case you’re in, and paste the two `findstr` results once you’re done.

okay so ill try running the same code again then?

No, rerunning won’t change anything. The Task 3 code isn’t in your laptop’s file, so it’ll behave exactly like before. First find out where your edits went.

Step 1: Check the Codespace

[CODESPACE]

```
git status

grep -c "exhausted" game/game_engine.py
```

- `grep` prints a number above 0: your edits are in the Codespace and just never got to your laptop. Go to Step 2.
- `grep` prints 0: you never made the Task 3 edits anywhere. Skip to Step 3.

Step 2: Edits are in the Codespace, so push them to the laptop

[CODESPACE]

```
git add game/game_engine.py
git commit -m "Task 3: add surge warning and exhaustion indicator"

git push
```

[CMD]

```
git pull

findstr /C:"exhausted" game\game_engine.py
```

You should see about 8 lines. Then run the balance fix below.

Step 3: No edits anywhere, so do them fresh on the laptop

[CMD]

```
code game\game_engine.py
```

Then go through Task 3 Steps 1-6 from my earlier message. Press Ctrl+S at the end. Make one change: in Step 2, replace `self.stamina - 2.0` with `self.stamina - 5.0`. It must say 2 replacements.

Balance fix (Step 2 path only)

[EDITOR] Ctrl+H, find `self.stamina - 3.0`, replace with `self.stamina - 5.0`, then Replace All. It must say 2 replacements. Save.

Verify, then test

[CMD]

```
findstr /C:"exhausted" game\game_engine.py
findstr /C:"stamina - 5.0" game\game_engine.py

python main.py
```

Press R, then mash Left/Right fast. After about 3 seconds you should get a flashing red bar and “EXHAUSTED!”.

Tell me what the Step 1 `grep` printed, and I’ll know which path you’re on.

Images hidden when shared

Good progress: the Task 3 code is in your file. The crash is a different problem, and it’s a clear one.

```
AttributeError: 'GameEngine' object has no attribute 'update_ai_phase'
```

`update()` calls `update_ai_phase()`, but the engine can’t find that method, although `set_ai_phase` exists. It worked when you committed Task 2, so something got deleted or knocked out of line during the Task 3 edits. The usual causes are the `def update_ai_phase` block being deleted along with a replaced chunk, or its indentation changing so it’s no longer a method.

Step 1: See the method layout

[CMD]

```
findstr /N /C:"def " game\game_engine.py
```

Paste the full output. Every `def` should start with exactly 4 spaces, and you should see these, in this order: `__init__`, `handle_event`, `set_ai_phase`, `update_ai_phase`, `update`, `reset`, `render`.

Step 2: Fix

Case A: `update_ai_phase` is missing from the list. [EDITOR] Paste this directly above `def update(self) :`, with a blank line before and after:

python

```
def update_ai_phase(self):
    elapsed = pygame.time.get_ticks() - self.phase_start
    if self.ai_phase == "BUILDING" and elapsed >= self.build_ms:
        self.set_ai_phase("SURGE")
    elif self.ai_phase == "SURGE" and elapsed >= self.surge_ms:
        self.set_ai_phase("COOLDOWN")
    elif self.ai_phase == "COOLDOWN" and elapsed >= self.cooldown_ms:

        self.set_ai_phase("BUILDING")
```

Case B: it's there but has extra spaces, or sits inside another method. Select the whole block and fix it so `def` is at exactly 4 spaces and its body at 8.

Step 3: Fix the stamina drain (still undone)

Your `findstr "stamina - 5.0"` printed nothing, so Step 2 of Task 3 isn't applied. [EDITOR] Press `Ctrl+H`, find `self.stamina - 2.0`, replace with `self.stamina - 5.0`, and click `Replace All`. It must say 2 replacements. Save with `Ctrl+S`.

Step 4: Verify and run

[CMD]

```
findstr /C:"stamina - 5.0" game\game_engine.py
```

```
python main.py
```

- The first command should show 2 lines.

- The game should start with no traceback. Press R, then mash. You should see the orange surge banner at about 4 seconds, and the flashing red bar plus “EXHAUSTED!” after about 3 seconds of fast mashing.

Paste the `findstr /N "def "` output (and any new traceback) and I’ll tell you exactly what to change.

Images hidden when shared

no i dont have update ai bro, its just update and i have it

You’re right, and that output confirms it. `update_ai_phase` got deleted somewhere during the Task 3 edits. The list has `set_ai_phase` (line 57), then jumps straight to `update` (line 63). Nothing is wrong with your `update`, it just calls a method that no longer exists. Put it back:

Step 1: Open the file

[CMD]

```
code game\game_engine.py
```

Step 2: Paste the missing method

[EDITOR] Press Ctrl+G, type 63, and press Enter. That jumps to `def update(self):`. Click at the very start of that line (before the `def`), and paste this block, then press Enter twice to keep a blank line before `def update`:

python

```
def update_ai_phase(self):
    elapsed = pygame.time.get_ticks() - self.phase_start
    if self.ai_phase == "BUILDING" and elapsed >= self.build_ms:
        self.set_ai_phase("SURGE")
    elif self.ai_phase == "SURGE" and elapsed >= self.surge_ms:
        self.set_ai_phase("COOLDOWN")
    elif self.ai_phase == "COOLDOWN" and elapsed >= self.cooldown_ms:
```

```
self.set_ai_phase("BUILDING")
```

Make sure `def update_ai_phase` lines up with `def set_ai_phase` above it (same indent) and there's one blank line between the methods. Don't press Tab. If VS Code's auto-indent adds extra spaces, fix them by hand.

Step 3: Drain fix, still pending

[EDITOR] Press Ctrl+H, find `self.stamina - 2.0`, replace with `self.stamina - 5.0`, and click Replace All. It must say 2 replacements. If it says 0, the drain was already changed, so tell me what the line currently says.

Save with Ctrl+S.

Step 4: Verify

[CMD]

```
findstr /N /C:"def " game\game_engine.py  
findstr /C:"stamina - 5.0" game\game_engine.py
```

- First: you should now see 7 defs, with `update_ai_phase` between `set_ai_phase` and `update`.
- Second: 2 lines.

Step 5: Run

[CMD]

```
python main.py
```

Press R, then mash Left/Right fast. After about 4 seconds you should see the orange “!! AI SURGE !!” banner, and after about 3 seconds of fast mashing, a flashing red bar plus “EXHAUSTED!”.

Paste any new error exactly, or tell me what you saw if it runs.

i still see 6 defs what

Six defs means the file on disk hasn't changed. `findstr` reads the saved file, so the usual cause is that the paste is in your editor but wasn't saved. Check these in order:

Check 1: Is it saved?

[EDITOR] Look at the file's tab at the top. A white dot instead of an X means unsaved changes. Press Ctrl+S, then rerun:

[CMD]

```
findstr /N /C:"def " game\game_engine.py
```

Check 2: Are you editing the right file?

You have used both the Codespace and your laptop. If the tab you pasted into is the Codespace browser tab, your laptop file never changed. Look at the window title:

- It should say your laptop's folder, not "Codespaces".
- The safest way to be sure is to close everything and reopen from [CMD]:

```
code game\game_engine.py
```

Check 3: Did the paste actually land?

[CMD]

```
findstr /N /C:"update_ai_phase" game\game_engine.py
```

You should see **2 lines**: the `def update_ai_phase` and the `self.update_ai_phase()` call inside `update`.

- Only the call shows: the paste isn't in the saved file. Go back to Check 1.
- Both show but the def count is still 6: the paste has wrong indentation (extra spaces or inside another method), so `findstr "def "` doesn't list it. Paste the output of the command above and tell me which line numbers it shows.

If it's still stuck

[CMD] Run this and paste me the output:

```
findstr /N /C:"def " /C:"update_ai_phase" game\game_engine.py
```

Don't retype the method. We just need to see what's actually in the saved file.

Images hidden when shared

That output explains it. Two clues:

1. `'code' is not recognized`: VS Code isn't set up on your laptop's command line, so `code game\game_engine.py` never opened anything. That was my mistake, I kept giving you a command that doesn't work on your machine.
2. `findstr` shows only the *call* on line 67, no `def update_ai_phase`. The paste never reached the laptop file. Wherever you pasted it (probably the Codespace tab), it's not this file.

So we edit with Notepad, which is guaranteed to open the laptop's file.

Step 1: Open it

[CMD]

```
notepad game\game_engine.py
```

Step 2: Paste the missing method

[NOTEPAD]

1. Press Ctrl+F, search for `def update(self) :`, and click Find Next. Close the find box with Esc.
2. Press Home so the cursor is at the very start of that line.
3. Paste this block exactly as is (it ends with an empty line):

python

```
def update_ai_phase(self):
    elapsed = pygame.time.get_ticks() - self.phase_start
    if self.ai_phase == "BUILDING" and elapsed >= self.build_ms:
        self.set_ai_phase("SURGE")
    elif self.ai_phase == "SURGE" and elapsed >= self.surge_ms:
        self.set_ai_phase("COOLDOWN")
    elif self.ai_phase == "COOLDOWN" and elapsed >= self.cooldown_ms:

        self.set_ai_phase("BUILDING")
```

4. Check that `def update_ai_phase` lines up with `def set_ai_phase` above it, with one blank line between the methods.

Step 3: Drain fix, still pending

[NOTEPAD] Press Ctrl+H. Find `self.stamina - 2.0`, replace with `self.stamina - 5.0`, then click Replace All. Then press Ctrl+S to save.

Step 4: Verify

[CMD]

```
findstr /N /C:"def " game\game_engine.py
```

```
findstr /C:"stamina - 5.0" game\game_engine.py
```

- First: you should now see 7 defs, with `update_ai_phase` right before `update`.
- Second: 2 lines.

Step 5: Run

[CMD]

```
python main.py
```

Press R, then mash fast. The orange “!! AI SURGE !!” banner should appear after about 4 seconds, and a flashing red bar plus “EXHAUSTED!” after about 3 seconds of hard mashing.

If the def count is still 6, paste the output of `findstr /N /C:"update_ai_phase" game\game_engine.py` and I’ll look at it. Tell me what you see.

yes both are working thank god

Good. Before committing, run a quick check so Task 3 goes in working.

Step 1: Test it

[CMD]

```
python main.py
```

Press R, then:

You do	You should see
Wait ~4 s	Flashing orange “!! AI SURGE !!” banner plus orange border for ~1.5 s
Mash 8+ presses/sec for ~3 s	Bar flashes red, “EXHAUSTED!” appears, your keys stop
Stop for ~2 s	Warning clears, keys work again

Check all three, and tell me if any of them doesn’t show up. Don’t commit a half-working task.

Step 2: Commit (only once all three work)

[CMD]

```
git status
git add game/game_engine.py
git commit -m "Task 3: add surge warning and exhaustion indicator"

git push
```

`git status` should list only `game/game_engine.py` as modified (plus the untracked `__pycache__`, which you ignore). After the push, `git log --oneline -4` should show Task 3 on top, above Task 2 and Task 1.

Task 4 preview

What it adds: if you press a key within about 1 second after the AI's surge ends (the moment cooldown begins), you trigger a counter-surge. That gives you an instant stamina refill and double push strength for ~3 seconds.

What it touches:

- A new “counter-surge” timer in `__init__` and `reset()`
- `set_ai_phase`: records when cooldown begins
- `handle_event`: checks for a press inside the window, and doubles your push while the bonus is active
- `render()`: a green “COUNTER-SURGE!” banner so the video shows it

Tell me how the test goes and I'll walk you through Task 4 in the same step-by-step format.

Images hidden when shared

Two things in that screenshot, and neither is serious.

1. The commit failed. The `git add` worked, but git doesn't know who you are on this laptop ("Author identity unknown"). So there is no Task 3 commit yet. The `git push` you typed at the bottom has nothing to push, so don't run it yet.

2. The debug prints are back. The top of your screenshot shows `AI phase: SURGE / COOLDOWN / BUILDING` again, so the `print(...)` line in `set_ai_phase` is in your file. It probably came back with the Task 3 edits. Delete it before committing, because that line already cost you a cleanup commit once.

Step 1: Set your git identity

Use the email linked to your ryonahh GitHub account, so the commits count on that profile.

[CMD]

```
git config --global user.name "ryonahh"

git config --global user.email "your-ryonahh-email@example.com"
```

Replace the email with the real one. To keep your email private, GitHub also gives you a `...@users.noreply.github.com` address under Settings, then Emails.

Step 2: Remove the debug print

[CMD]

```
findstr /N /C:"print(" game\game_engine.py
```

- Nothing printed: it's already gone, skip to Step 3.
- A line like `print("AI phase:", phase) # TEMP...` shows: open the file and delete that whole line.

[CMD]

```
notepad game\game_engine.py
```

[NOTEPAD] Press Ctrl+F, search `TEMP`, delete the entire line, then Ctrl+S.

[CMD] Re-run the `findstr` above. It should print nothing now.

Step 3: Quick retest

[CMD]

```
python main.py
```

The terminal should now stay silent, with no `AI phase:` lines. In the game, check the three things from before: the surge banner at about 4 s, red bar plus “EXHAUSTED!” after fast mashing, and recovery after a pause. Tell me if any of the three is missing.

Step 4: Commit and push

[CMD]

```
git add game/game_engine.py
git commit -m "Task 3: add surge warning and exhaustion indicator"

git push
```

The first push from this laptop may open a browser sign-in. Log in as ryonahh. Then:

[CMD]

```
git log --oneline -4
```

You should see Task 3 on top, then Task 2 (remove debug print), Task 2 (add cycle), Task 1. Paste it to me and we start Task 4.

Images hidden when shared

its fine

Both pushes went through. The log shows Task 1, Task 2 (two commits) and Task 3 on top, and the debug print is gone. Task 3 appears twice with the same message: one was your earlier Codespace push, one is today's final fix. That's harmless, so leave it. Use distinct messages from here on.

Task 4: Counter-surge bonus

What it does: if you press a key within about 1.2 s of the AI's surge ending (the start of COOLDOWN), you get a one-time bonus. You get an instant +30 stamina, 4x stamina regen for 3 s, and double push (8.4 instead of 4.2) for 3 s. A green banner shows it.

Everything is in `game/game_engine.py`. Edit in Notepad, since `code` doesn't work on your laptop.

[CMD]

```
notepad game\game_engine.py
```

Edit 1: New variables in `__init__`

[NOTEPAD] Ctrl+F, search `self.recover_at = 30`. Paste directly under it, same indent (8 spaces):

```
python
```

```
    self.counter_window_ms = 1200
    self.counter_bonus_ms = 3000
    self.counter_active = False
    self.counter_used = False

    self.counter_start = 0
```

Edit 2: Allow one bonus per cooldown

[NOTEPAD] Ctrl+F, search `self.ai_phase = phase`. It matches once, inside `set_ai_phase`. Paste directly under that line, same indent:

python

```
self.counter_used = False
```

Edit 3: Double push

[NOTEPAD] Ctrl+H. Find `self.arm_position -= 4.2`, replace with `self.arm_position -= self.push_strength()`, then Replace All. It must say 2 replacements.

Edit 4: Add the `push_strength` method

[NOTEPAD] Ctrl+F, search `def update(self) :`. Press Home so the cursor is at the start of that line. Paste this block, ending with an empty line:

python

```
def push_strength(self):
    now = pygame.time.get_ticks()
    if (not self.counter_active and not self.counter_used
        and self.ai_phase == "COOLDOWN"
        and now - self.phase_start <= self.counter_window_ms):
        self.counter_active = True
        self.counter_used = True
        self.counter_start = now
        self.stamina = min(self.max_stamina, self.stamina + 30.0)

    return 8.4 if self.counter_active else 4.2
```

Check that `def push_strength` lines up with the other `defs` (4 spaces) and there's one blank line between methods.

Edit 5: Bonus expiry, in `update()`

[NOTEPAD] Ctrl+F, search `self.update_ai_phase()`. Paste directly under that line, same indent:

python

```
        if self.counter_active and pygame.time.get_ticks() -  
        self.counter_start > self.counter_bonus_ms:  
  
            self.counter_active = False
```

Edit 6: Faster regen during the bonus

[NOTEPAD] Ctrl+H. Find `self.stamina + 0.2`, replace with `self.stamina + (0.8 if self.counter_active else 0.2)`, then Replace All. It must say 1 replacement. If it says 0, tell me what your regen line currently says.

Edit 7: Reset it

[NOTEPAD] Ctrl+F, search `def reset(self) :`. Paste directly under that line, 8 spaces:

python

```
        self.counter_active = False
```

Edit 8: Banner in `render()`

[NOTEPAD] Ctrl+F, search `if self.game_state == "GAME_OVER" :`. Click at the start of that line and paste above it, then leave a blank line:

python

```
        if self.counter_active and self.game_state == "PLAYING":  
            cs = self.font_big.render("COUNTER-SURGE! x2", True, (80, 240,  
120))  
            screen.blit(cs, (self.width // 2 - cs.get_width() // 2, 510))  
  
            pygame.draw.rect(screen, (80, 240, 120), pygame.Rect(0, 0,  
self.width, self.height), width=6)
```

Press Ctrl+S to save.

Verify

[CMD]

```
findstr /N /C:"def " game\game_engine.py
findstr /C:"push_strength" game\game_engine.py

findstr /C:"print(" game\game_engine.py
```

- First: 8 defs, with `push_strength` right before update.
- Second: 3 lines (the def and 2 calls).
- Third: nothing printed.

Test

[CMD]

```
python main.py
```

Press R, then mash alternating Left/Right, but don't go all-out during the surge. You can't press while exhausted, and you can only trigger the bonus by pressing right after the surge ends.

You see	Meaning
Orange banner (surge), then a green "COUNTER-SURGE! x2" banner almost immediately	Working
Hands move about twice as fast for 3 s	Double push working
Stamina bar refills noticeably	Recovery boost working
No green banner	You were exhausted or stopped pressing when the surge ended, so try again

Problem	Cause
Traceback with push_strength	Edit 4 missing or mis-indented
AttributeError: counter_active	Edit 1 missing
Banner appears and never leaves	Edit 5 missing

Commit (only after the test works)

[CMD]

```
git add game/game_engine.py
git commit -m "Task 4: add counter-surge bonus"
git push
```

```
git log --oneline -3
```

Paste the log, or any error, and tell me what you saw. Then we record the after video and package the Lab-4 folder.

yep im done

“Done” could mean edited, tested, or committed, and I can’t tell which. Run these checks before we call it.

Step 1: Verify the file

[CMD]

```
findstr /N /C:"def " game\game_engine.py
findstr /C:"push_strength" game\game_engine.py
findstr /C:"print(" game\game_engine.py
```

```
git status
```

- First: 8 defs, with `push_strength` before update.
- Second: 3 lines.
- Third: nothing.
- `git status`: only `game_engine.py` modified (plus the `__pycache__` folder you ignore), or clean if you've already committed.

Step 2: Did the game actually show it?

[CMD]

```
python main.py
```

Press R, then mash. When the orange surge banner ends, keep pressing. You need all of these:

- A green “COUNTER-SURGE! x2” banner right after the surge
- The hands moving visibly faster for ~3 s
- The stamina bar jumping up

Tell me honestly if any of the three is missing, and don't commit yet in that case.

Step 3: Commit (only if all three worked)

[CMD]

```
git add game/game_engine.py
git commit -m "Task 4: add counter-surge bonus"
git push
```

```
git log --oneline -3
```

Paste the log so I can confirm Task 4 is on top.

Next: the “after” video

Once Task 4 is pushed:

1. Start recording (Win+Alt+R).
2. Run `python main.py`, click the window, press R.
3. In 10 seconds, show: mashing pulls the hands to the left, then the orange surge banner, then the green counter-surge banner, then (if you can fit it) the EXHAUSTED! red bar.
4. Stop recording, rename to `after.mp4`.

The exhaustion state is the hardest to fit into 10 s, because winning quickly ends the game. A good approach is to mash hard at the start to drain stamina, then pause, then press right as the surge ends. Do a couple of practice runs first.

Question I still need answered

Does your course want the Lab-4 folder in this repo or a separate lab repo? Ask your TA or a classmate before the final push, because it decides where the videos and the chat export go.

Tell me what Steps 1 and 2 showed.

done with testing as well and it works thank you

Good. Last stretch: commit, record, package.

Step 1: Commit Task 4

[CMD]

```
git status
```

- `game_engine.py` shows as modified: you haven't committed yet. Run:

```
git add game/game_engine.py
```

```
git commit -m "Task 4: add counter-surge bonus"
```

```
git push
```

- “working tree clean” (apart from `__pycache__`): already committed. Check it was pushed with `git log --oneline -3`. Task 4 should be on top, and `origin/main` should be on the same line.

Step 2: Record `after.mp4`

1. Press Win+Alt+R to start recording.
2. [CMD] `python main.py`, click the window, press R.
3. Aim to show these, in about 10 seconds:
 - Mash fast and the hands go left (bug fixed)
 - Orange !! AI SURGE !! banner
 - Green COUNTER-SURGE! x2 right after it
 - The red EXHAUSTED! bar, if you can fit it. Mash hard early to drain stamina.
4. Press Win+Alt+R to stop. The file lands in `Videos\Captures`. Rename it to `after.mp4`.

Practice a couple of runs first. If you win too fast to show everything, don't mash all-out in the first seconds. It's fine to let the computer lead for a bit.

Step 3: Build the Lab-4 folder

[CMD] from your project folder:

```
mkdir Lab-4
copy game\game_engine.py Lab-4\
copy main.py Lab-4\
```

Then drag `before.mp4` and `after.mp4` into the `Lab-4` folder in File Explorer.

Step 4: Export this chat (do this last)

Chat export has to include everything, so do it after all steps are done. Ctrl+P on this page, then Save as PDF. Check that it captured the whole conversation, not just the visible part. Save it as `chat_history.pdf` in `Lab-4`.

Step 5: Push

[CMD]

```
git add Lab-4
git commit -m "Add Lab-4 deliverables"

git push
```

Videos are large. If the push is rejected as too big (GitHub's limit is 100 MB per file), tell me and we'll compress them.

One open question

Does your course want the Lab-4 folder in this repo or a separate lab repo? Ask your TA or a classmate before you do Step 5. Also, the “chat link” in the README says the full chat history, so copy this conversation's share link too if your platform offers one.

Tell me what the `git status` or `git log` in Step 1 showed.

Images hidden when shared

not getting chat history

The preview shows one blank page because Claude's chat scrolls inside a container, so the browser's print only captures the empty page shell. It's a known annoyance, and there are three ways around it. Try them in order.

Option 1: Share link, then print the shared page

This also covers the README's “Chat/LLM used page link”.

1. [BROWSER] On this chat, click Share (top right), then create the link and copy it.
2. Open the link in a new tab.
3. Press Ctrl+P on that page. Shared pages are laid out as normal documents, so they usually print with the whole conversation.
4. Check the page count in the preview. It should be well above 1 page. If it is, save as PDF.

Option 2: Copy-paste into Word

If the shared page still prints blank:

1. [BROWSER] Click in the chat, press Ctrl+A, then Ctrl+C. If it only grabs part of the chat, scroll to the top first and repeat.
2. Open Word, paste with Ctrl+V, then File, then Save As, and pick PDF (or keep the .docx). The handout accepts either doc or pdf.
3. Screenshots won't carry over, but all the text and code will, and that's what shows your prompts and iterations.

Option 3: Last resort

Scroll through the chat in chunks and print only the visible parts, or paste it in sections. It's slow, so only use it if both options above fail.

Timing

This chat is still going, so export last, after your final messages. A share link is a snapshot from when you create it. Messages sent after that may not appear, so create or refresh the link at the very end.

Do this after your commit and videos are done. Tell me which option worked, or what you see if the shared page also prints blank.

This is the read-only copy that you shared of your chat.